JAIDEV S RAM

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Aspiring Computational Scientist

Passionate and results-driven MSc student in Data Analytics and Computational Science, with expertise in machine learning (ML), deep learning (DL), and natural language processing (NLP). Experienced in applying advanced computational techniques to solve complex problems in areas such as nonlinear dynamics, chaos theory, and neuroscience. Skilled in data analysis, mathematical modeling, and scientific computing, seeking to leverage these skills in interdisciplinary research and development projects.

EXPERIENCE

Research and Development Intern **Sep 2023 - Present**

Digital University, Kerala

- Collaborated with a cross-functional team to develop and implement an OCR-based solution for extracting and verifying candidate experience certificates.
- Leveraged **Tesseract OCR** to extract text from scanned documents and convert them into structured data formats, enabling seamless analysis and verification.
- Applied **data preprocessing techniques** to clean and standardize extracted text, ensuring the accuracy and integrity of the data for downstream analysis.
- Conducted **data quality checks** and implemented error-correction protocols to minimize inaccuracies and optimize the data extraction pipeline.

EDUCATION

Digital University, Kerala **Sep 2023 - Present**

MSc. Data Analytics and Computational Science

Kirori Mal College, University of Delhi **May 2022**

BSc (Hons.) Physics

Cumulative Grade Point Average (CGPA) 8.041

Government Boys Higher Secondary School, Nemmara **Mar 2019**

Higher Secondary (Kerala Board)

Passed with 96.6%

TECHNICAL SKILLS

Programming languages:

- **Python:** Proficient in solving complex problems using libraries like NumPy, Pandas, SciKit-Learn, and Matplotlib. Hands-on experience with natural language processing (NLP) using spaCy for text analysis.
- **Scilab:** Applied to solve mathematical and computational problems in fields like quantum mechanics, statistical physics, and mathematical physics.
- **R:** Basic proficiency with data analysis, statistical modeling, and visualization.
- **C++:** Foundational knowledge of object-oriented programming and algorithmic problem-solving.

Software & Tools:

- **Latex:** Used for preparing professional academic documents, reports, and research papers.
- **Git/GitHub:** Version control for collaborative research and project management.

Operating Systems:

- **Windows** and **Linux**: Proficient in navigating and utilizing both environments for software development and scientific computing tasks.

PROJECTS

Detecting Signs of Depression from Social Media Text:

- Developed a machine learning model to detect early signs of depression by analyzing text data from social media platforms. The project involved data preprocessing, feature extraction, and classification using various machine learning algorithms.
- **Technologies:** Python (NLP), Scikit-learn, Pandas, SpaCy

Drug-Target Prediction for Neurodegenerative Diseases:

- Leveraged computational techniques to predict potential drug-target interactions for neurodegenerative diseases like Alzheimer's and Parkinson's. Focused on dataset preprocessing, feature engineering, and building predictive models using classification algorithms.
- **Technologies:** Python (Machine Learning), Scikit-learn, Pandas

PUBLICATIONS

1. Impact of Lévy noise on spiral waves in a lattice of Chialvo neuron map

This study explores how external noise with unique statistical properties can impact spiral waves in a lattice of Chialvo neurons. We investigated how network parameters like coupling strength and range, as well as noise parameters, influence the dynamics and existence of spiral waves.

DOI: https://doi.org/10.1016/j.chaos.2024.115759

2. Spatiotemporal patterns in a 2D lattice of Hindmarsh–Rose neurons induced by high-amplitude pulses

Investigated the impact of high-amplitude Gaussian pulses on the emergence of novel stable regimes within a Hindmarsh–Rose (HR) neuronal network.

DOI: https://doi.org/10.1016/j.chaos.2024.115613

EXTRA-CURRICULAR ACTIVITIES

Junior Red Cross (JRC) Cadet:

Active participant in the Junior Red Cross, where I engaged in various social and humanitarian activities, including health awareness campaigns, first aid workshops, and disaster management initiatives.